Middle East Technical University
Department of Electrical and Electronics Engineering

EE201 – Circuit Theory I

Lecture 7: Circuit Graphs, the Incidence Matrix and Tellegen's Theorem

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Colour code used in these notes and in the slides			
current i	voltage v, e	power	nodes

1. Interconnection equations

A circuit with b branches has $2b$ unknowns: a voltage v_k and a current i_k for each branch k. The $2b$ equations that fix them come in two groups.

Group	How many	What they describe
element equations	b	*what* each element is: $v = Ri$, a source value, a dependent source, ...
interconnection equations	b	*how* the elements are connected: $n-1$ KCL and $b-n+1$ KVL equations

The interconnection equations do not depend on the elements at all. Replace a resistor by a capacitor or by a source, keep the same wiring, and KCL and KVL do not change. So, to write KCL and KVL, we may forget the elements and keep only the wiring. What is left is the **graph** of the circuit.

This lecture builds the graph of a circuit and writes *all* of its KCL and KVL equations with one matrix. Tellegen's theorem then follows in one line. The same bookkeeping is how a circuit simulator stores a circuit: a list of elements, each with the nodes it joins.

2. The graph of a circuit

Circuit graph

The **graph** of a circuit is the circuit with every two-terminal element replaced by a line segment, called a **branch**. The points where branches meet are the **nodes**. Each branch gets a number and an arrow.

The graph records only *which nodes each branch joins*, and in which direction it is taken. Lengths, angles and the shape of the drawing carry no information.

2.1. Procedure

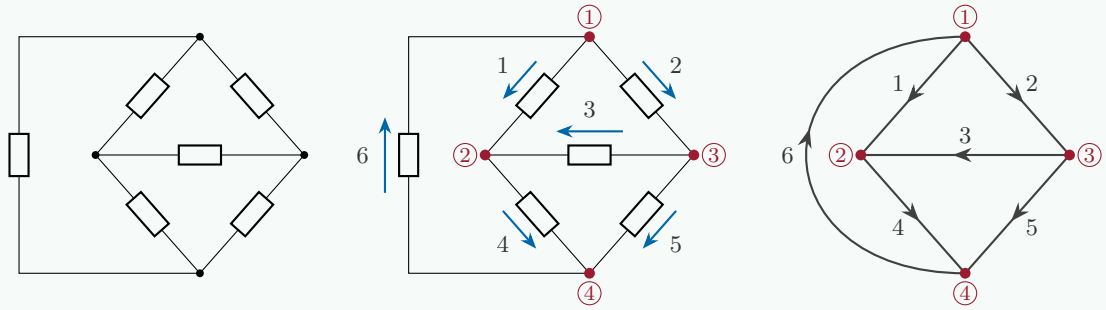
1. Label each branch: $1, 2, \ldots, b$.
2. Assign a direction to each branch. The choice is **arbitrary**.
3. Label each node: ①, ②, ..., ⓝ.
4. Draw the graph: a dot for each node and an arrow for each branch.

The arrow on branch k is the **reference direction** of its current i_k. As before we use associated reference directions, so the branch voltage v_k is measured with $+$ at the tail of the arrow and $-$ at its head. The arrow is only a reference: if the current flows the other way, i_k is negative.

2.2. Example

Example 7.1 — the graph of a bridge circuit

The bridge circuit below has six two-terminal elements, drawn as boxes because their type does not matter here. Find its graph.



(a) bridge circuit (b) steps 1–3 (c) step 4: the graph

Steps 1–3 label the six branches, give each one an arrow (blue: the current reference), and label the four nodes. Step 4 keeps only the dots and the arrows. The bridge graph has

$$\text{nodes } \{①, ②, ③, ④\}, \quad n = 4; \qquad \text{branches } \{1, 2, 3, 4, 5, 6\}, \quad b = 6.$$

Branch 6 is drawn curved only to keep it clear of the others; a straight arrow from ④ to ① would be the same graph.

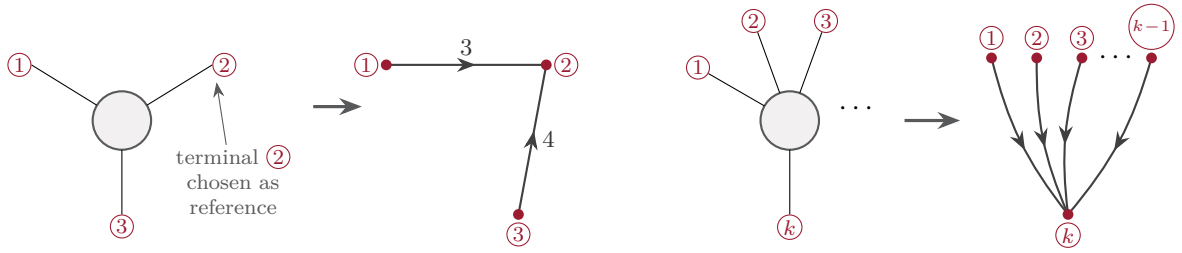
3. Elements with more than two terminals

A two-terminal element becomes one branch. Some elements have more terminals: a transistor has three, an integrated circuit can have many. Such an element cannot be one line segment.

A k-terminal element in the graph

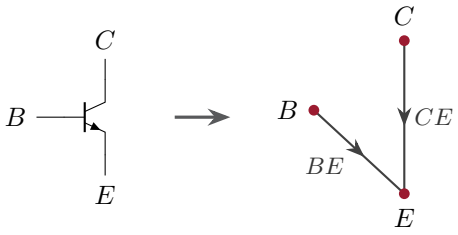
For an element with $k \geq 3$ terminals, choose one terminal as its **reference** (the choice is arbitrary). Then draw $k-1$ branches, one from each of the other terminals to the reference terminal.

Why $k-1$ branches, not k? A closed surface around the element is crossed by k terminal currents, and KCL on that surface says they sum to zero. Only $k-1$ of them are free; the last one follows. The voltages work the same way: the $k-1$ voltages measured from the reference terminal fix every other voltage between terminals, by KVL. So $k-1$ branches, each with a current and a voltage, describe the element completely.



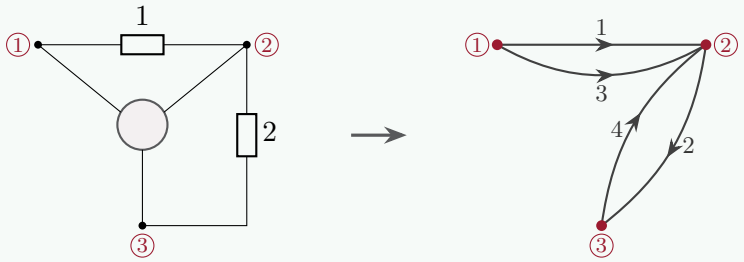
a 3-terminal element $k-1=2$ branches a k-terminal element $k-1$ branches

A transistor is the most common case. With its emitter E as the reference terminal it becomes two branches: base to emitter, and collector to emitter.



Example 7.2 — a circuit with a 3-terminal element

In the circuit below, element 1 joins nodes ① and ②, element 2 joins ② and ③, and a 3-terminal element touches all three nodes. Find the graph.



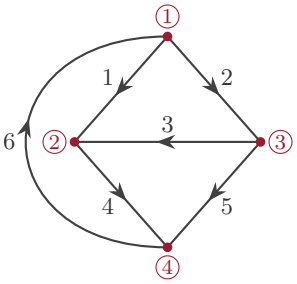
the circuit — its graph: $n = 3$, $b = 4$

Elements 1 and 2 give one branch each. For the 3-terminal element choose the terminal at ② as reference; it gives branch 3 (from ① to ②) and branch 4 (from ③ to ②). Branches 1 and 3 join the same two nodes; that is allowed, and so is the pair 2 and 4.

A dependent source needs nothing new: it is already drawn as two branches, the branch that senses the controlling variable and the branch of the source.

4. KCL on a circuit graph

Every branch current flows along its own arrow. At each node, count a current **leaving** the node with + and a current **entering** it with −; KCL says the sum is zero. For the bridge graph in Example 7.1:



$$
\begin{aligned}
① &: \quad i_1 + i_2 - i_6 = 0 \\
② &: \quad -i_1 - i_3 + i_4 = 0 \\
③ &: \quad -i_2 + i_3 + i_5 = 0 \\
④ &: \quad -i_4 - i_5 + i_6 = 0
\end{aligned}
$$

Nothing about the elements was used: the four equations come from the graph alone. Two circuits with the same graph have the same KCL equations, whatever their elements are.

5. The incidence matrix

The four equations above contain only the coefficients $+1$, -1 and 0. Collect them in a matrix, and collect the currents in a vector:

$$\begin{array}{c} \\ ① \\ ② \\ ③ \\ ④ \end{array}\!\!\begin{array}{c} \begin{array}{cccccc} 1 & 2 & 3 & 4 & 5 & 6 \end{array} \\ \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & -1 \\ -1 & 0 & -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 & -1 & 1 \end{bmatrix} \end{array} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \\ i_4 \\ i_5 \\ i_6 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \qquad \Longleftrightarrow \qquad A_a\,\mathbf{i} = \mathbf{0}.$$

Rows are labelled by nodes, columns by branches. A matrix that stores the graph like this is a **circuit matrix**. There are several; the first and most basic is this one.

Incidence matrix

For a graph with n nodes and b branches, the **incidence matrix** A_a is the $n \times b$ matrix with entries (j: node, k: branch)

$$a_{jk} = \begin{cases} +1 & \text{if branch } k \text{ leaves node } j, \\ -1 & \text{if branch } k \text{ enters node } j, \\ 0 & \text{otherwise.} \end{cases}$$

With the **branch current vector** $\mathbf{i} = [\, i_1 \; i_2 \; \cdots \; i_b \,]^T$, KCL at every node of the circuit is the single equation $A_a\,\mathbf{i} = \mathbf{0}$.

The subscript a stands for *all*: every node has a row. The matrix can be read in two directions.

- **A row** is a node. Its nonzero entries are the branches that touch that node, and the row times $\mathbf{i}$ is the KCL equation at that node.
- **A column** is a branch. It has exactly one $+1$ (the node the branch leaves) and exactly one -1 (the node it enters). Column 3 above has $+1$ in row ③ and -1 in row ②: branch 3 goes from ③ to ②.

The columns hold all of the graph: from A_a alone one can redraw the graph, arrows included. Reversing the arrow of a branch flips the sign of its column, and nothing else.

6. Review: linear dependence

Linear dependence

Let $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k$ be vectors of the same size. If there are real numbers $\alpha_1, \alpha_2, \ldots, \alpha_k$, **not all zero**, with

$$\alpha_1\mathbf{v}_1 + \alpha_2\mathbf{v}_2 + \cdots + \alpha_k\mathbf{v}_k = \mathbf{0},$$

the vectors are **linearly dependent**. Otherwise — if the only such numbers are $\alpha_1 = \cdots = \alpha_k = 0$ — they are **linearly independent**.

If the vectors are dependent, at least one of them is a combination of the others: pick an $\alpha_m \neq 0$ and solve for $\mathbf{v}_m$.

Example 7.3 — dependent or not?

$\begin{bmatrix}1\\0\end{bmatrix}, \begin{bmatrix}0\\-1\end{bmatrix}$ are independent; $\mathbf{v}_1 = \begin{bmatrix}1\\2\\3\end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix}4\\5\\6\end{bmatrix}, \mathbf{v}_3 = \begin{bmatrix}7\\8\\9\end{bmatrix}$ are dependent.

In the first pair, $\alpha_1[1\ 0]^T + \alpha_2[0\ -1]^T = [\alpha_1\ \ -\alpha_2]^T$, which is zero only if $\alpha_1 = \alpha_2 = 0$. In the second set, $\mathbf{v}_1 - 2\mathbf{v}_2 + \mathbf{v}_3 = \mathbf{0}$ (check one entry: $1 - 10 + 9 = 0$ in the middle row), so $\mathbf{v}_3 = 2\mathbf{v}_2 - \mathbf{v}_1$. Any nonzero multiple works as well, for example $3\mathbf{v}_1 - 6\mathbf{v}_2 + 3\mathbf{v}_3 = \mathbf{0}$.

For a set of equations the meaning is direct. If the rows of a matrix M are dependent, one equation of $M\mathbf{x} = \mathbf{0}$ follows from the others and adds no information. The largest number of independent rows is the **rank** of M. When all rows are independent, M has **full row rank**.

7. The reduced incidence matrix

Look at the incidence matrix of the bridge. Column 3 (branch 3, from ③ to ②) has $+1$ in row ③, -1 in row ②, and zeros elsewhere. Every column is like this: one $+1$ and one -1. So adding all four rows gives zero in every column:

$$A_a = \begin{array}{c} \\ ① \\ ② \\ ③ \\ ④ \\ \text{sum of rows:} \end{array} \begin{array}{c} \begin{array}{cccccc} 1 & 2 & 3 & 4 & 5 & 6 \end{array} \\ \left[\begin{array}{cccccc} 1 & 1 & 0 & 0 & 0 & -1 \\ -1 & 0 & -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 & -1 & 1 \end{array}\right] \\ \begin{array}{cccccc} 0 & 0 & 0 & 0 & 0 & 0 \end{array} \end{array}$$

The same holds for any graph:

$$\text{row } ① + \text{row } ② + \cdots + \text{row } ⓝ = [0\ 0\ \cdots\ 0].$$

This is the definition of linear dependence with every $\alpha = 1$. The rows of A_a are **always** linearly dependent, and one of the n KCL equations is redundant — the fact met earlier, now in matrix form.

Reduced incidence matrix

Choose one node as the **reference node** (or **ground**). Delete its row from A_a. The $(n-1)\times b$ matrix that remains is the **reduced incidence matrix**, written A.

The choice of the reference node is arbitrary. It is the same node that will serve as the common reference for node voltages.

Example 7.4 — the reduced incidence matrix of the bridge

Take ④ as the reference node and delete its row:

$$A_a = \left[\begin{array}{cccccc} 1 & 1 & 0 & 0 & 0 & -1 \\ -1 & 0 & -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 & -1 & 1 \end{array}\right] \xrightarrow{\text{delete row ④}} A = \begin{array}{c} \\ ① \\ ② \\ ③ \end{array} \begin{array}{c} \begin{array}{cccccc} 1 & 2 & 3 & 4 & 5 & 6 \end{array} \\ \left[\begin{array}{cccccc} 1 & 1 & 0 & 0 & 0 & -1 \\ -1 & 0 & -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 & 1 & 0 \end{array}\right] \end{array}.$$

A is 3×6. Are its rows independent? Let α_1 (row ①) $+\alpha_2$ (row ②) $+\alpha_3$ (row ③) $= \mathbf{0}$. Columns 4, 5 and 6 (boxed) have only one nonzero entry each, so each of them fixes one α:

$$\text{column 4: } \alpha_2 = 0, \qquad \text{column 5: } \alpha_3 = 0, \qquad \text{column 6: } -\alpha_1 = 0.$$

All three α's are zero, so the rows of A are independent.

Deleting the row loses nothing.

- **No equation is lost.** The rows of A_a add up to zero. So the deleted row is the negative of the sum of the kept rows. For the bridge:

  $$\text{row } ④ = -(\text{row } ① + \text{row } ② + \text{row } ③): \qquad \begin{bmatrix} 0 & 0 & 0 & -1 & -1 & 1 \end{bmatrix} = -\begin{bmatrix} 0 & 0 & 0 & 1 & 1 & -1 \end{bmatrix}.$$

  The same holds for the equations. KCL at ④ is the negative of the sum of KCL at ①, ② and ③:

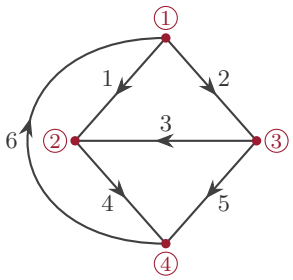
  $$-i_4 - i_5 + i_6 = -\big[(i_1 + i_2 - i_6) + (-i_1 - i_3 + i_4) + (-i_2 + i_3 + i_5)\big].$$

  If KCL holds at ①, ②, ③ ($A\,\mathbf{i} = \mathbf{0}$), the right side is zero, so KCL at ④ holds too ($A_a\,\mathbf{i} = \mathbf{0}$).
- **No part of the graph is lost.** Each column of A_a sums to zero, so the deleted row can be rebuilt from A. A column of A with a **single nonzero** entry is a branch that touches the reference node.

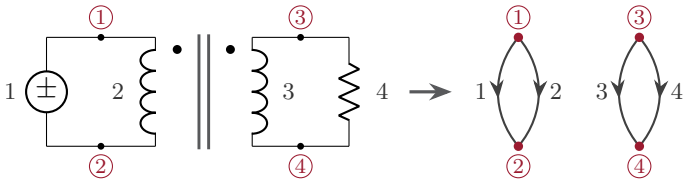
8. Connected graphs

Connected graph

A graph is **connected** if any two nodes are joined by a path of branches, ignoring the arrow directions. Otherwise it falls into separate pieces.



bridge graph: connected



a transformer circuit and its graph: two pieces

The bridge graph is connected. The transformer circuit is not: the windings are coupled only through the core, so no branch joins the two sides. (An ideal transformer is a two-port: one branch per winding.) Other circuits split the same way, for example an isolation amplifier or an optocoupler: a dependent source senses in one part and acts in the other.

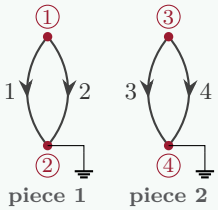
Connected graph: the rows of A are independent

If the graph is connected, the $n-1$ rows of A are linearly independent: $\operatorname{rank} A = n - 1$.

8.1. A graph in separate pieces

Treat each piece as a graph of its own. Each piece is connected, so it gets its own incidence matrix, its own reference node and its own reduced incidence matrix. Then stack the reduced matrices into one.

Example 7.5 — the transformer circuit



incidence matrix **reference** **reduced**

piece 1 $\begin{matrix} & 1 & 2 \\ ① & 1 & 1 \\ ② & 1 & 1 \end{matrix}$ delete row ② $A_1 = ① \begin{bmatrix} 1 & 1 \end{bmatrix}$

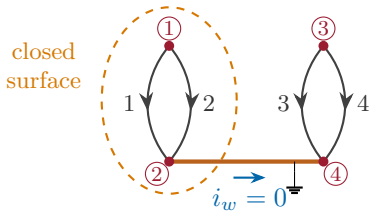
piece 2 $\begin{matrix} & 3 & 4 \\ ③ & 1 & 1 \\ ④ & 1 & 1 \end{matrix}$ delete row ④ $A_2 = ③ \begin{bmatrix} 1 & 1 \end{bmatrix}$

Stack: $A = \begin{bmatrix} A_1 & 0 \\ 0 & A_2 \end{bmatrix} = \begin{matrix} ① \\ ③ \end{matrix} \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}$ $\quad i_1 + i_2 = 0, \quad i_3 + i_4 = 0.$

Two independent KCL equations: $n - p = 4 - 2 = 2$.

With p separate pieces there are p reference nodes, one per piece, and A has $n - p$ independent rows.

Corollary: joining the pieces. Join the two reference nodes ② and ④ of the transformer graph with a wire. They become one node. Three things to check: the current in the wire, the branch voltages, and the incidence matrix.



The wire carries no current. Let i_w flow through the wire from ② to ④. Branches 1 and 2 go from ① to ②, so KCL at ① and at ② gives

$$①: \quad i_1 + i_2 = 0, \qquad ②: \quad -i_1 - i_2 + i_w = 0 \qquad \Longrightarrow \qquad i_w = i_1 + i_2 = 0.$$

This is KCL on the closed surface around piece 1: only the wire crosses it.

No branch voltage changes. Every branch still lies inside its own piece. The wire only gives the two pieces a common reference.

The incidence matrix stays the same. The joined graph is connected. Its nodes are ①, ③ and the joined node ② ≡ ④, so $n = 3$ and $b = 4$. Branches 3 and 4 now go from ③ to the joined node. Column by column, with the joined node as reference:

$$A_a = \begin{matrix} ① \\ ② \\ ③ \end{matrix} \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix} \equiv ④ \quad \xrightarrow{\text{delete row ②}} \quad A = \begin{matrix} ① \\ ③ \end{matrix} \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix} = \begin{bmatrix} A_1 & 0 \\ 0 & A_2 \end{bmatrix}$$

This is exactly the stacked A of Example 7.5. The KCL equation at the joined node, $-(i_1 + i_2) - (i_3 + i_4) = 0$, is the deleted row: it follows from the other two.

In general. Joining the p reference nodes changes only their rows, and these rows are deleted anyway. All other rows stay as they were. The joined graph has $n - p + 1$ nodes, so A again has $n - p$ rows. A graph in separate pieces can always be made connected this way, and nothing changes. In EE201 graphs are connected unless stated otherwise.

9. KCL in matrix form

KCL in matrix form

For a circuit whose graph is connected, with n nodes and b branches,

$\boxed{A\,\mathbf{i} = \mathbf{0}}$ $\quad A \in \mathbb{R}^{(n-1)\times b}$: reduced incidence matrix, $\quad \mathbf{i} \in \mathbb{R}^{b}$: branch current vector.

These are $n-1$ **independent** KCL equations, and every other KCL equation of the circuit follows from them. For a graph in p separate pieces, build A piece by piece; it has $n-p$ rows.

Example 7.6 — KCL for the circuit with a 3-terminal element

Write A_a, A and the KCL equations for the graph of Example 7.2, with ③ as the reference node.

The arrows are: branch 1 from ① to ②, branch 2 from ② to ③, branch 3 from ① to ②, branch 4 from ③ to ②. Column by column:

$$A_a = \begin{array}{c} \\ ① \\ ② \\ ③ \end{array}\!\!\begin{array}{c} \begin{array}{cccc} 1 & 2 & 3 & 4 \end{array} \\ \begin{bmatrix} 1 & 0 & 1 & 0 \\ -1 & 1 & -1 & -1 \\ 0 & -1 & 0 & 1 \end{bmatrix} \end{array} \qquad A = \begin{bmatrix} 1 & 0 & 1 & 0 \\ -1 & 1 & -1 & -1 \end{bmatrix},$$

$$A\,\mathbf{i} = \mathbf{0}: \qquad i_1 + i_3 = 0, \qquad -i_1 + i_2 - i_3 - i_4 = 0.$$

Columns 1 and 3 are equal because branches 1 and 3 join the same nodes in the same direction. The first equation says the current leaving ① through element 1 returns through the element's terminal at ①. The reference terminal of the element (at ②) and the reference node of the graph (③) are two separate choices.

10. Branch voltages and node voltages

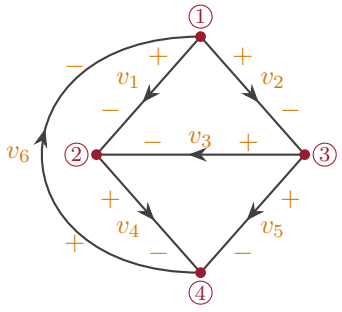
The incidence matrix also gives KVL. For that we need two kinds of voltage.

Branch voltages and node voltages

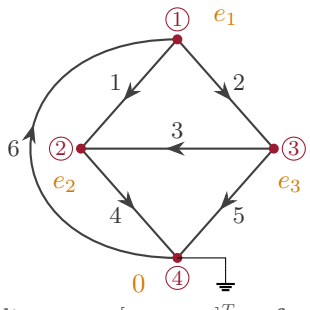
Branch voltage v_k: the voltage across branch k, + at the tail of the arrow. So i_k enters at + (associated reference).



Node voltage e_k: the voltage of node k with respect to the reference node. The reference node has $e = 0$.



branch voltages $\mathbf{v} = [\,v_1 \;\cdots\; v_6\,]^T$

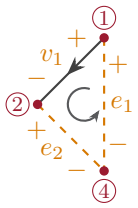


node voltages $\mathbf{e} = [\,e_1\;\, e_2\;\, e_3\,]^T$, reference ④

$b = 6$ branch voltages, but only $n - 1 = 3$ node voltages. Each node voltage e_k has + at node k and − at the reference node ④.

11. KVL on a circuit graph

By KVL, every branch voltage can be written in terms of node voltages. Branch 1 of the bridge goes from ① to ②. Branch 1 and the node voltages e_1, e_2 form a closed loop ① → ② → ④ → ①. KVL around it:



$$v_1 + e_2 - e_1 = 0 \quad \Rightarrow \quad v_1 = e_1 - e_2.$$

In general, a branch from node p to node q has $v_k = e_p - e_q$: **tail minus head**. If one end is the reference node, its node voltage is 0. For all six branches of the bridge:

$$\left.\begin{aligned} v_1 &= e_1 - e_2 \\ v_2 &= e_1 - e_3 \\ v_3 &= e_3 - e_2 \\ v_4 &= e_2 \\ v_5 &= e_3 \\ v_6 &= -e_1 \end{aligned}\right\} \quad \Longrightarrow \quad \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \\ v_6 \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & -1 & 0 \\ 1 & 0 & -1 \\ 0 & -1 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & 0 & 0 \end{bmatrix}}_{A^T} \begin{bmatrix} e_1 \\ e_2 \\ e_3 \end{bmatrix}.$$

The matrix is A^T, the transpose of A from Example 7.4: its row k is column k of A. For example, row 1 is $[\,1 \;\; -1 \;\; 0\,]$, and column 1 of A is $[\,1 \;\; -1 \;\; 0\,]^T$.

KVL in matrix form

$\boxed{\mathbf{v} = A^T \mathbf{e}}$ $\quad \mathbf{v} \in \mathbb{R}^b$: branch voltage vector, $\quad \mathbf{e} \in \mathbb{R}^{n-1}$: node voltage vector.

The same matrix A gives KCL ($A\mathbf{i} = \mathbf{0}$) and KVL ($\mathbf{v} = A^T\mathbf{e}$).

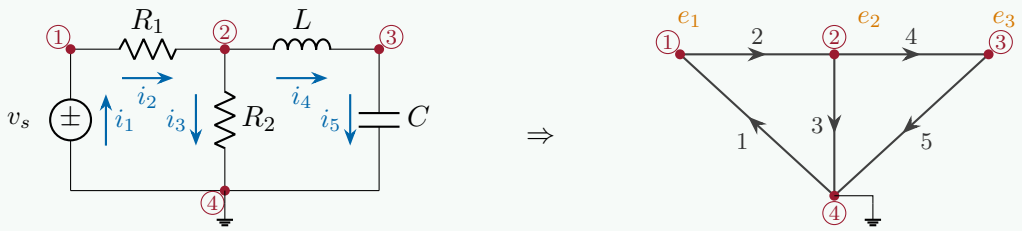
Remarks.

- Once the $n-1$ node voltages are known, all b branch voltages are known.
- KVL puts **no constraint** on the node voltages: any $\mathbf{e}$ gives branch voltages that satisfy KVL around every loop. Around a loop the differences cancel. For the loop of branches 1, 3, 2: $v_1 - v_3 - v_2 = (e_1 - e_2) - (e_3 - e_2) - (e_1 - e_3) = 0$.

So the node voltages are a good choice of unknowns: there are only $n-1$ of them, they are free, and KVL is then satisfied automatically. This is the idea behind node analysis.

Example 7.7 — branch voltages from node voltages

Draw the graph of the circuit below, find A (reference ④), and write every branch voltage in terms of the node voltages.



Each element is one branch, with the current reference shown. Then

$$A = \begin{bmatrix} -1 & 1 & 0 & 0 & 0 \\ 0 & -1 & 1 & 1 & 0 \\ 0 & 0 & 0 & -1 & 1 \end{bmatrix}, \qquad \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \end{bmatrix} = \begin{bmatrix} -1 & 0 & 0 \\ 1 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \\ e_3 \end{bmatrix} = \begin{bmatrix} -e_1 \\ e_1 - e_2 \\ e_2 \\ e_2 - e_3 \\ e_3 \end{bmatrix}.$$

Each row is tail minus head. Branch 1 goes from the reference ④ to ①, so $v_1 = 0 - e_1$. The element types play no role: a source, resistors, an inductor or a capacitor — only the graph matters.

12. Tellegen's theorem

KCL and KVL together give a short but powerful result.

Tellegen's theorem

In any circuit, with associated reference directions,

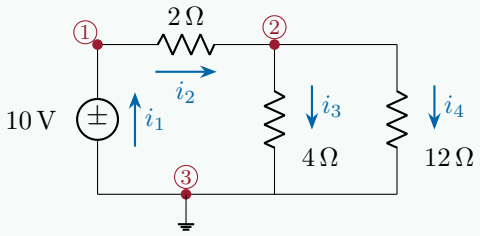
$$\mathbf{i}^T \mathbf{v} = \sum_{k=1}^{b} v_k \, i_k = 0 \qquad \text{at every instant } t.$$

$v_k i_k$ is the power absorbed by branch k. So the theorem says the total power absorbed by all branches is zero: the power delivered by some branches is absorbed by the others. This is **conservation of power**.

The proof uses KCL and KVL only, never the element equations. So the theorem holds even more generally, for two different circuits with the same graph. The general form and its proof are in the appendix.

Example 7.8 — checking Tellegen's theorem

Find the voltage and the current of every branch with node voltages, and check that $\sum_k v_k i_k = 0$. Each branch has the current reference shown; its voltage has + at the tail of that arrow. Reference node: ③.



1. Node voltages. The source fixes $e_1 = 10\,\text{V}$; e_2 is the only unknown.

2. KCL at ②, with each current written by Ohm's law in terms of node voltages ($-i_2 + i_3 + i_4 = 0$):

$$-\frac{e_1 - e_2}{2} + \frac{e_2}{4} + \frac{e_2}{12} = 0 \quad \xrightarrow{\times 12} \quad -6(10 - e_2) + 3e_2 + e_2 = 0 \quad \Rightarrow \quad e_2 = 6\,\text{V}.$$

3. Branch voltages, $\mathbf{v} = A^T\mathbf{e}$ (tail minus head): $v_1 = 0 - e_1 = -10\,\text{V}$, $v_2 = e_1 - e_2 = 4\,\text{V}$, $v_3 = v_4 = e_2 = 6\,\text{V}$.

4. Branch currents, Ohm's law: $i_2 = 4/2 = 2\,\text{A}$, $i_3 = 6/4 = 1.5\,\text{A}$, $i_4 = 6/12 = 0.5\,\text{A}$. KCL at ①: $-i_1 + i_2 = 0$, so $i_1 = i_2 = 2\,\text{A}$.

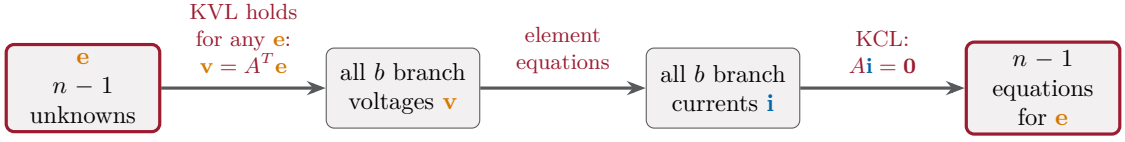
5. Check.

branch	element	v_k (V)	i_k (A)	$v_k i_k$ (W)
1	source	−10	2	−20
2	2 Ω	4	2	8
3	4 Ω	6	1.5	9
4	12 Ω	6	0.5	3
			sum	0

Branch 1 goes from ③ to ①, so the + of v_1 is at the bottom and $v_1 = -10\,\text{V}$: $v_1 i_1 = -20\,\text{W}$. The source delivers 20 W, and the resistors absorb $8 + 9 + 3 = 20\,\text{W}$.

13. Looking ahead: why node voltages?

A circuit has $2b$ unknowns, every v_k and every i_k, and $2b$ equations: b element equations, $n-1$ KCL equations and $b-n+1$ KVL equations. The results of this lecture show that a much smaller set of unknowns is enough.



- **KVL.** Take the $n-1$ node voltages as the unknowns and write every branch voltage as $\mathbf{v} = A^T\mathbf{e}$. As shown in Section 11, this satisfies KVL around every loop for *any* $\mathbf{e}$. So no KVL equation has to be written: the $b - n + 1$ KVL equations hold automatically.
- **Element equations.** These are still used: they give each branch current from its voltage, hence in terms of $\mathbf{e}$. For a resistor, $i_k = v_k/R_k = (e_{\text{tail}} - e_{\text{head}})/R_k$.
- **KCL.** Put these currents into $A\mathbf{i} = \mathbf{0}$. This leaves $n-1$ equations, exactly as many as the unknowns $\mathbf{e}$.

Solve them for $\mathbf{e}$, and every other voltage and current follows. So $2b$ unknowns become $n-1$.

Always $b > n-1$

Joining n nodes takes at least $n-1$ branches. With exactly $n-1$ there is no loop. Then each branch is the only link between two parts of the graph, and KCL on a surface around one part makes its current zero (as for the wire in Section 8.1). So a circuit that carries any current has a loop, and

$$n-1 \;<\; b \;<\; 2b.$$

Each branch beyond the first $n-1$ closes a loop and adds one KVL equation: hence $b-n+1$ of them.

Example 7.8 was a small case of it, with the single unknown e_2. This is **node analysis**, the subject of a coming lecture.

Summary

- The $2b$ equations of a circuit split into b **element equations** and b **interconnection equations** (KCL, KVL). The second group depends only on the wiring, that is, on the **graph**.
- The graph: number the branches, give each an arbitrary arrow (the current reference), number the nodes, and keep only dots and arrows. A k-terminal element becomes $k-1$ branches to a reference terminal.
- The **incidence matrix** A_a $(n \times b)$: $+1$ where a branch leaves a node, -1 where it enters. KCL at all nodes is $A_a\mathbf{i} = \mathbf{0}$. Each column has one $+1$ and one -1, so the rows sum to zero and are dependent.
- The **reduced incidence matrix** A $((n-1) \times b)$: A_a without the reference node's row. For a **connected** graph its rows are independent, and $A\mathbf{i} = \mathbf{0}$ gives exactly the $n-1$ independent KCL equations. For a graph in separate pieces: one reference node per piece, then stack.
- **Node voltages** $\mathbf{e}$ ($n-1$ of them, measured from the reference). KVL: $\mathbf{v} = A^T\mathbf{e}$, each branch voltage is tail minus head. KVL puts no constraint on $\mathbf{e}$.
- **Tellegen's theorem**: in any circuit $\mathbf{i}^T\mathbf{v} = \sum v_k i_k = 0$ — conservation of power, from KCL and KVL alone. (General form and proof: appendix.)
- **Why**: with the $n-1$ node voltages as unknowns, KVL holds automatically (no KVL equation to write), and KCL gives just enough equations — the idea of node analysis.

Next lecture

Planar graphs and meshes: the **mesh matrix** M, KVL as $M\mathbf{v} = \mathbf{0}$, and mesh currents with $\mathbf{i} = M^T\mathbf{i}_M$ — the counterpart of $\mathbf{v} = A^T\mathbf{e}$.

Appendix

A.1 General Tellegen's theorem

The theorem of Section 12 needs only two facts: $\mathbf{i}$ satisfies KCL and $\mathbf{v}$ satisfies KVL, both on the same graph. Nothing requires them to come from the same circuit.

General Tellegen's theorem

Take two circuits with the **same graph**, with any elements. Let $\mathbf{i}$ be the branch currents of circuit 1 and $\mathbf{v}$ the branch voltages of circuit 2, with associated reference directions. Then

$$\mathbf{i}^T\mathbf{v} = \sum_{k=1}^{b} v_k\, i_k = 0.$$

Here $v_k i_k$ is no longer a power: the voltage and the current belong to different circuits. With circuit 1 = circuit 2 it is conservation of power.

Example A.1 — the general form on the bridge graph

Take any branch currents that satisfy KCL on the bridge graph, and any node voltages — they do not have to come from the same circuit:

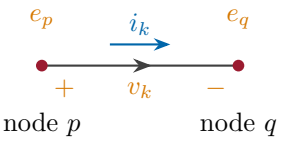
$$\mathbf{i} = [1 \;\; 2 \;\; 1 \;\; 2 \;\; 1 \;\; 3]^T \text{ A}, \qquad \mathbf{e} = [5 \;\; 2 \;\; 3]^T \text{ V}.$$

KCL holds: $1+2-3=0$, $-1-1+2=0$, $-2+1+1=0$. The branch voltages are $\mathbf{v} = A^T\mathbf{e} = [3 \;\; 2 \;\; 1 \;\; 2 \;\; 3 \;\; -5]^T$ V. Then

$$\mathbf{i}^T\mathbf{v} = 1\cdot 3 + 2\cdot 2 + 1\cdot 1 + 2\cdot 2 + 1\cdot 3 + 3\cdot(-5) = 15 - 15 = 0.$$

A.2 Proof of the general Tellegen's theorem

Let $\mathbf{i}$ be the branch currents of circuit 1: they satisfy KCL, $A\mathbf{i} = \mathbf{0}$. Let $\mathbf{v}$ be the branch voltages of circuit 2, and $\mathbf{e}$ its node voltages: they satisfy KVL, $\mathbf{v} = A^T\mathbf{e}$. Both use the same A, because the graph is the same. Take branch k from node p to node q.



By KVL in circuit 2, its voltage is tail minus head, $v_k = e_p - e_q$. So the product $v_k i_k$ splits into two terms, one for each end:

$$v_k i_k = \underbrace{e_p i_k}_{i_k \text{ leaves } p} - \underbrace{e_q i_k}_{i_k \text{ enters } q}.$$

Add over all branches and collect the terms of each node n. Every branch that leaves n contributes $+e_n i_k$ and every branch that enters n contributes $-e_n i_k$:

$$\sum_k v_k i_k = \sum_n e_n \Big(\underbrace{\textstyle\sum \text{currents leaving } n - \sum \text{currents entering } n}_{=0 \text{ by KCL in circuit 1}} \Big) = 0.$$

(The reference node has $e = 0$, so it adds nothing.) For the bridge the brackets are exactly the three KCL equations of Section 4: $\sum_k v_k i_k = e_1(i_1+i_2-i_6) + e_2(-i_1-i_3+i_4) + e_3(-i_2+i_3+i_5)$. In matrix form the two steps are one line: $\mathbf{i}^T\mathbf{v} \overset{\text{KVL}}{=} \mathbf{i}^T A^T\mathbf{e} = (A\mathbf{i})^T\mathbf{e} \overset{\text{KCL}}{=} 0$. ■

The proof uses only KCL for $\mathbf{i}$ and KVL for $\mathbf{v}$, never the element equations. With circuit 1 = circuit 2 it gives the theorem of Section 12: conservation of power.